

Setting Up the Raspberry Pi for Internet of Things

Introduction

This document provides installation and configuration instructions for six Internet of Things (IoT) projects demonstrating a variety of techniques for controlling devices over the Internet. Figure 1 shows a conceptual view of the relationship between an Internet client and an Internet accessible device connected to the Raspberry Pi. Each project's user interface design maintains a "thin client" philosophy so that no special software need be installed on the client machine accessing the Internet thing. The projects range from simple control of an LED to that of maintaining a database and displaying recorded data from an altimeter sensor.

The instructions given in this document apply to most Linux distributions that run on the Raspberry Pi. The author has successfully run these projects on a Raspberry Pi 3 running the Raspberry Pi Trixie operating system. Network configuration and settings are not covered in this project hookup guide. It is assumed that the user has a means of connecting the Raspberry Pi to a local network and that the Internet is accessible from the local network. There are many possible ways of doing this. The networking aspect is beyond the scope of this document. Contact your local network administrator for instructions on how to connect your Raspberry Pi to your local network.

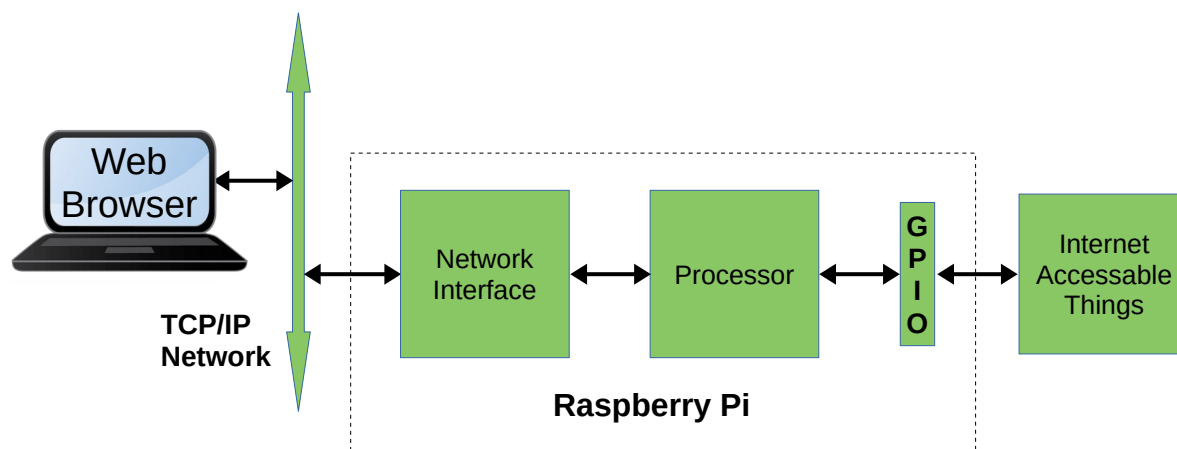


Figure 1. Overall conceptual view of Raspberry Pi internet accessible devices.

Raspberry Pi Internet of Things Overview

Hookup instructions and source code provided for the following six projects

Project	Description
LED	Turn on and off an LED
Push Button	Display on a web page the number of times a push button has been pushed

Servo	Control a servo by moving a slider on a web page
Fncgen	A simple, software defined function generator that demonstrates the uses of a digital to analog converter.
Oscilloscope	A software defined oscilloscope that demonstrates the use of an analog to digital converter.
Altimeter	An altimeter app that presents in charts and text format the data from an altitude sensor. Demonstrates the use of a round-robin database to store data and generate graphical charts.

Each project has its own folder in the Raspberry Pi Internet of Things (raspiot) repository. Each project folder has the following structure

- **docs** folder containing hookup instructions, schematics, and component data sheets
- **bin** folder containing executables that go in the **pi** user bin folder **/home/pi/bin**
- **html** folder containing html documents and php scripts that go in the **pi** user web document root folder **/home/pi/public_html**.

Inside the document root **public_html** folder a sub-folder should be created for each project that will be installed. For example, if you choose to install the LED project, you should create a folder **/home/pi/public_html/led** to contain all the files and sub-folders that are in the repository LED html folder. The files in each project bin folder always go into the common **/home/pi/bin** folder.

Hardware Hookup

Each Raspberry project has its own hookup guide in its documents folder. Refer to the project hookup guide for instructions on how to hookup the hardware components and install the software. Each project involves connecting devices to the Raspberry Pi GPIO. Figure 2 shows the location of the GPIO header on the Raspberry Pi printed circuit board and provides explanations of the GPIO pin outs. The hardware is best hooked up using standard bread-boarding components along with a Raspberry Pi GPIO breakout board. Figure 3 shows a photo of all six projects hooked up using only one breadboard.

Parts List

Project	Parts	Supplier / Part Number
LED	• LED (150-200mcd) • 220 Ohm ¼ Watt Resistor	Adafruit, Digikey, Sparkfun
Push Button	• Normally Open Single Pole Single Throw Push Button Switch (breadboard mountable) • 10 Kiloohm ¼ Watt Resistor	Adafruit, Digikey, Sparkfun # COM-09190

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Servo	• SpringRC SM-S2309S Micro Servo (or equivalent) • 1 Kiloohm ¼ Watt Resistor	Adafruit, Digikey, Sparkfun
Fncgen	• MCP4725 Digital to Analog Converter	Sparkfun # BOB-12918
Oscilloscope	• ADS1115 Analog to Digital Converter	Amazon.com
Altimeter	• MPL3115A2 Altimeter	Sparkfun # SEN-11084
Required for all projects	• Raspberry Pi 3 (or higher) • GPIO Breakout Connector • Solderless Breadboard • Jumper Wires	• Adafruit # 3055, Sparkfun # DEV-14643 • Adafruit # 2028, Sparkfun # BOB-13717 • Adafruit # 239, Sparkfun # PRT-12615 • Adafruit # 3314, Sparkfun # PRT-00124

Supplier Websites

Supplier	URL	Parts
Adafruit	https://www.adafruit.com	LED, Push Button, Resistors, ADS1115, Pi 3, Breakout Connector, Solderless Breadboard, Jumper Wires
Digikey	https://www.digikey.com/	LED, Push Button, Resistors, Pi 3, Breakout Connector, Solderless Breadboard, Jumper Wires
Sparkfun	https://sparkfun.com	LED, Push Button, Resistors, MCP4725, MPL3115, Pi 3, Breakout Connector, Solderless Breadboard, Jumper Wires
Amazon	https://amazon.com	ADS1115 Analog to Digital Converter

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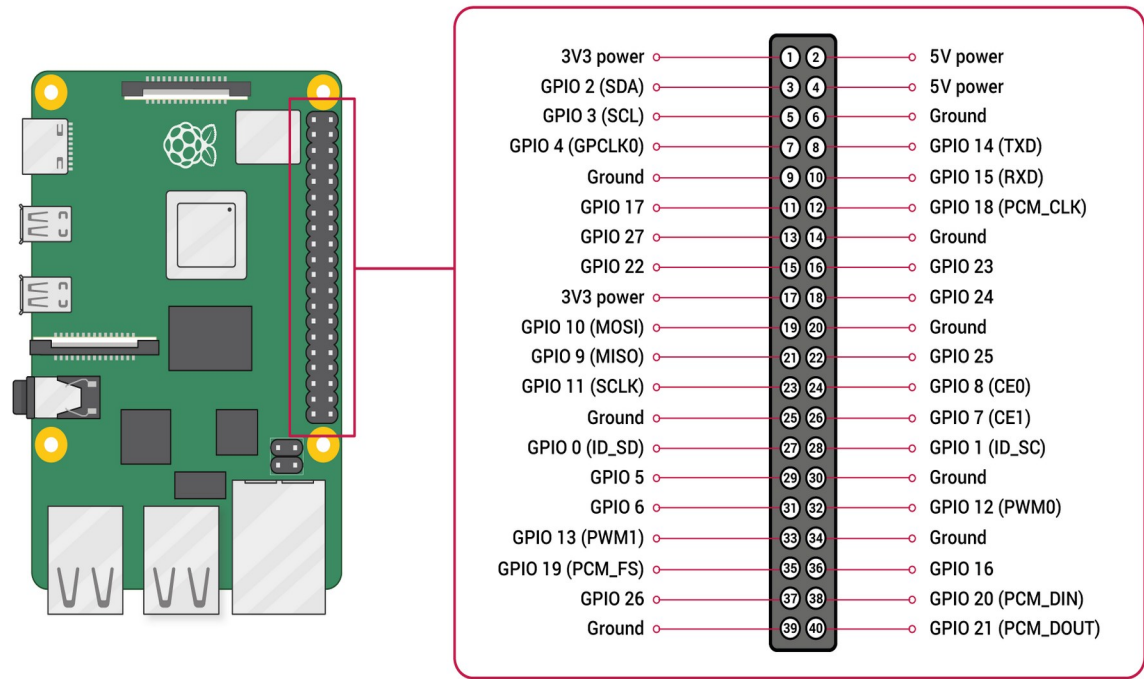


Figure 2. Diagram showing GPIO pin outs and location of GPIO header on Raspberry Pi PCB.

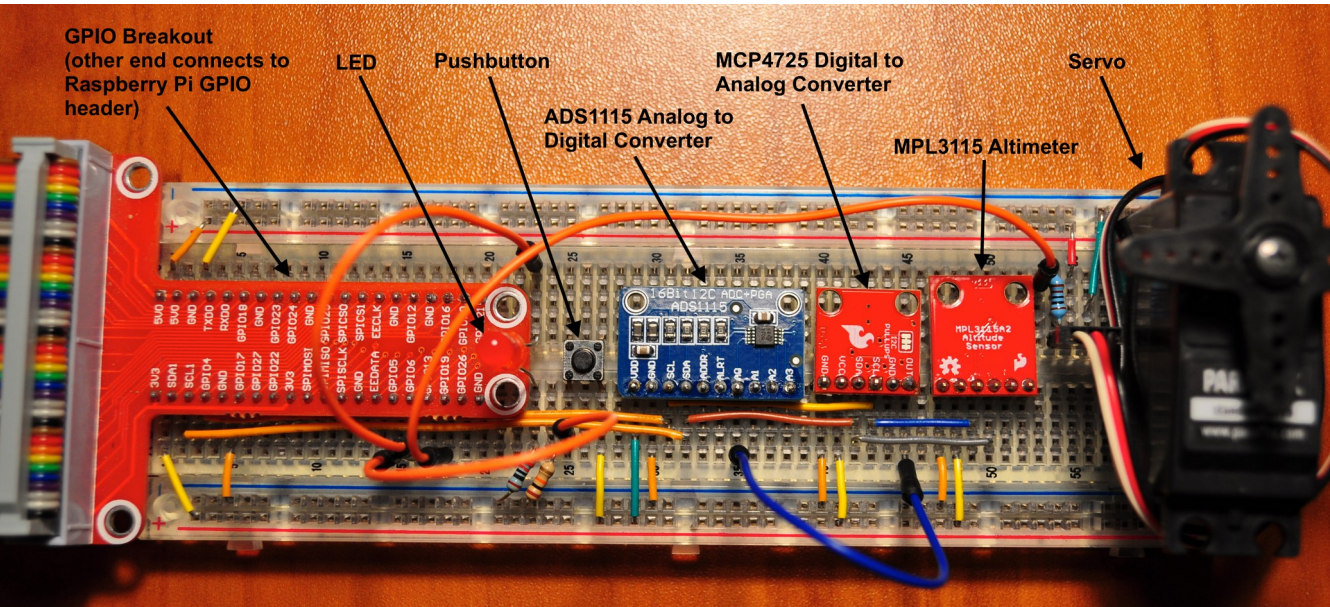


Figure 3. Photo of all six projects mounted on a single breadboard.

Server Software

Installing the Server Software

Dependencies:

The Raspberry Pi IoT apps should be installed on a Linux host which meets the following requirements:

- Raspbian Trixie OS (Debian GNU/Linux 13) or later version
- Apache 2 should be installed and configured to allow serving HTML documents from the user's public_html folder.
- PHP should be installed and configured to allow PHP scripts to run from the user's public_html folder.
- The Python I2C smbus library should be installed.
- Python 3.0 usually comes pre-installed in virtually all Linux distributions. Type "python3" at a command line prompt to verify Python 3 has been installed.

Software Inventory

The following software items need to be installed in the user documents root folder, e.g., **/home/pi/public_html**.

HTML folder:

- index.html
- apps.html
- favicon.ico
- static/bg_apps.jpg

The following item needs to be installed in the user **bin** folder, e.g., **/home/pi/public_html**.

bin folder:

- startup.sh

Getting the Raspberry Pi IoT Software

From the github repository:

1. On a computer connected to the Internet download the zip file from
<https://github.com/fractalxaos/raspiot/archive/refs/heads/main.zip>
2. From the downloaded file **raspiot-main.zip**, extract the **raspiot-main** folder to the desktop.
3. From the **raspiot-main** folder, copy files to the Raspberry Pi as directed in the following section and in the instructions for each individual project.

Installing on a Raspberry Pi

Figure 1 shows a block diagram of a Raspberry Pi connected to the local area network. The Pi can be either connected to the network via built-in WiFi adapter, or by Ethernet cable. Note that the following installation procedure assumes that the document root for the IoT HTML documents will be the user's `public_html` folder. Typically the full path name to this folder will be something like `/home/{user}/public_html`, where `{user}` is the name of the user account hosting the IoT project apps. The following steps will assume the apps are running under the `pi` user account. The `pi` account is the default account when Raspberry Pi OS is first installed on a Raspberry Pi.

1. Follow the instructions in the [Base System Installation Guide](#) to install and configure web services on your Raspberry Pi.
2. If necessary, turn on the I2C bus by running

```
sudo raspi-config
```

3. Select Interface Options.
4. Select I2C and select Yes. Select Ok and Finish.
5. Determine if the python I2C smbus library has been installed. Start the python interpreter by running

```
python3
```

At the python prompt enter

```
import smbus
```

If you get an error, then the smbus library has not been installed.

6. Similarly determine if the python psutil library has been installed. Type **Ctrl-D** to exit the python interpreter.
7. If necessary, install the I2C smbus and psutil python libraries by running

```
sudo apt update  
sudo apt install python3-smbus  
sudo apt install python3-psutil
```

8. If it doesn't already exist, create the file `/etc/rc.local`. Set the permissions to allow `rc.local` execution permissions. E.g.

```
sudo touch /etc/rc.local  
sudo chmod 750 /etc/rc.local
```

Add to `/etc/rc.local` the following lines

```
#!/bin/sh -e
(su - pi -c "bin/startup.sh")&
exit 0
```

9. If **/etc/rc.local** already exists, back up **/etc/rc.local** and add the line

```
(su - pi -c "bin/startup.sh")&
```

Be sure **exit 0** is the last line of the script.

Whenever the host boots up, the **startup.sh** script will run the start up commands.

10. If it doesn't already exist, use **mkdir** to create a folder **public_html** in the **pi** user's home folder. The working folder should look like **/home/pi/public_html**.
11. Move all the contents of the repository **raspiot-main/html/** folder to the **public_html** folder created in step 10. This includes the **static** subfolder and its contents.
12. If it does not already exist, use **mkdir** to create a folder named **bin** in the **pi** user's home folder. For example, the full path name should look like **/home/pi/bin**.
13. Move all the contents of the repository **raspiot-main/bin/** folder to the **bin** folder created in step 12.
14. Be sure that you grant **startup.sh** execute permissions by running

```
chmod u+x ~/bin/startup.sh
```

15. Follow the individual hookup instructions for each project you choose to install.

This completes installation of the server software.

References and Resources

The following resources describe how to configure a Raspberry Pi to be a server

- Using an SSH key pair:
<https://help.github.com/en/github/authenticating-to-github/generating-a-new-ssh-key-and-adding-it-to-the-ssh-agent>
- Reducing wear on the SD card:
<https://www.zdnet.com/article/raspberry-pi-extending-the-life-of-the-sd-card/>
- Installing a web server:
<https://raspberrypi.tips/en/raspberrypi-tutorials/raspberry-pi-web-server-apache-php-mariadb>
- Linux system administration and devops
<https://www.guru99.com/unix-linux-tutorial.html>

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<http://devopsbootcamp.osuosl.org/>
<http://lug.oregonstate.edu>

The following tutorials are useful for more in depth understanding of the server software:

- Javascript <http://www.w3schools.com/js/default.asp>
- PHP <http://www.w3schools.com/php/default.asp>
- HTML <http://www.w3schools.com/html/default.asp>
- Rrdtool <http://oss.oetiker.ch/rrdtool/>
- Python <http://greenteapress.com/thinkpython/thinkpython.html>